Electrical Properties of Materials

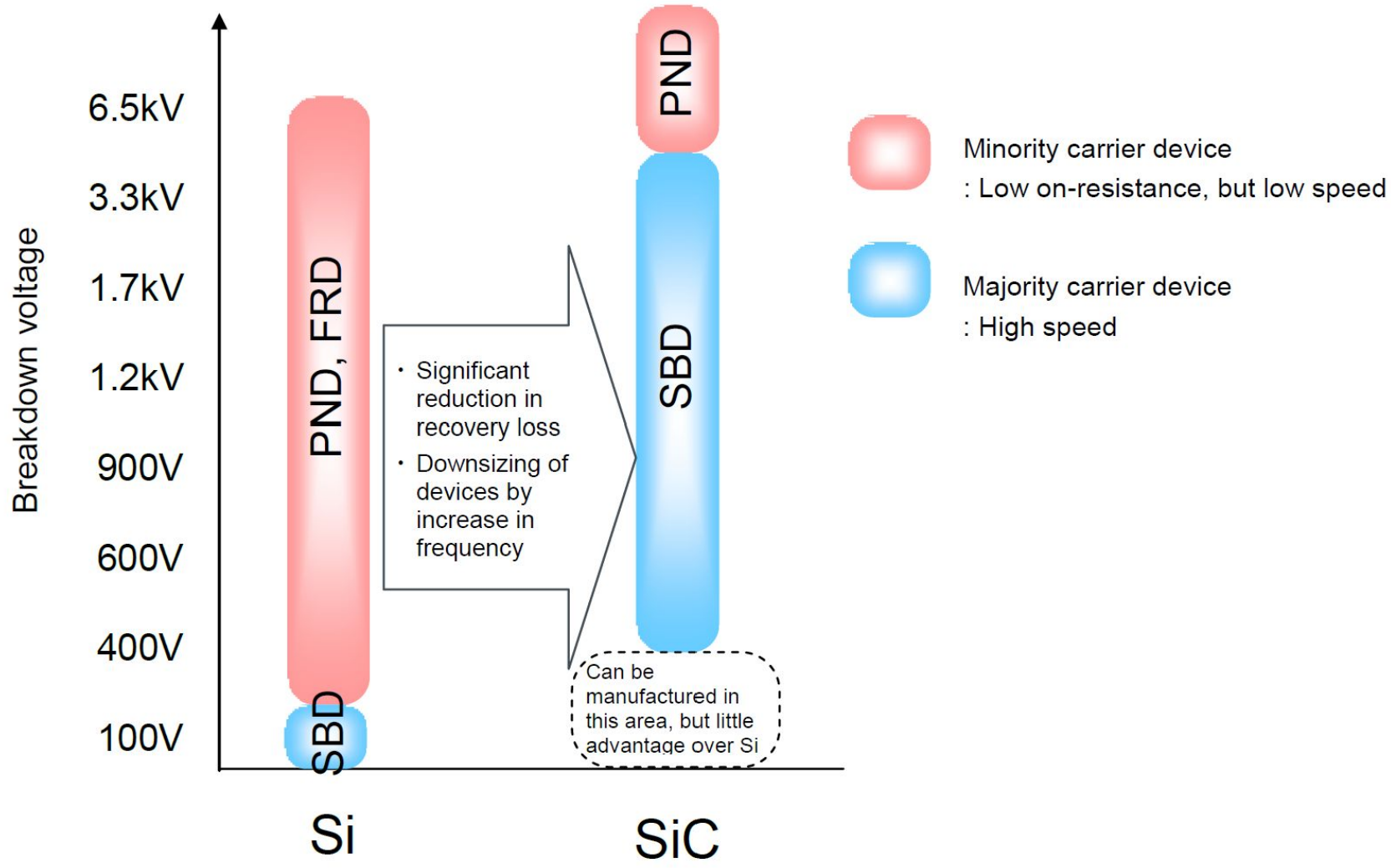
Properties	Si	4H-SiC	GaAs	GaN
Crystal Structure	Diamond	Hexagonal	Zincblende	Hexagonal
Energy Gap : E_G (eV)	1.12	3.26	1.43	3.5
Electron Mobility : μ_n (cm^2/Vs)	1400	900	8500	1250
Hole Mobility : μ_p (cm^2/Vs)	600	100	400	200
Breakdown Field : E_B (V/cm) X10^6	0.3	3	0.4	3
Thermal Conductivity (W/cm°C)	1.5	4.9	0.5	1.3
Saturation Drift Velocity : v_s (cm/s) X10^7	1	2.7	2	2.7
Relative Dielectric Constamt : ε_S	11.8	9.7	12.8	9.5
p, n Control	○	○	○	△
Thermal Oxide	○	○	×	×

Advantages

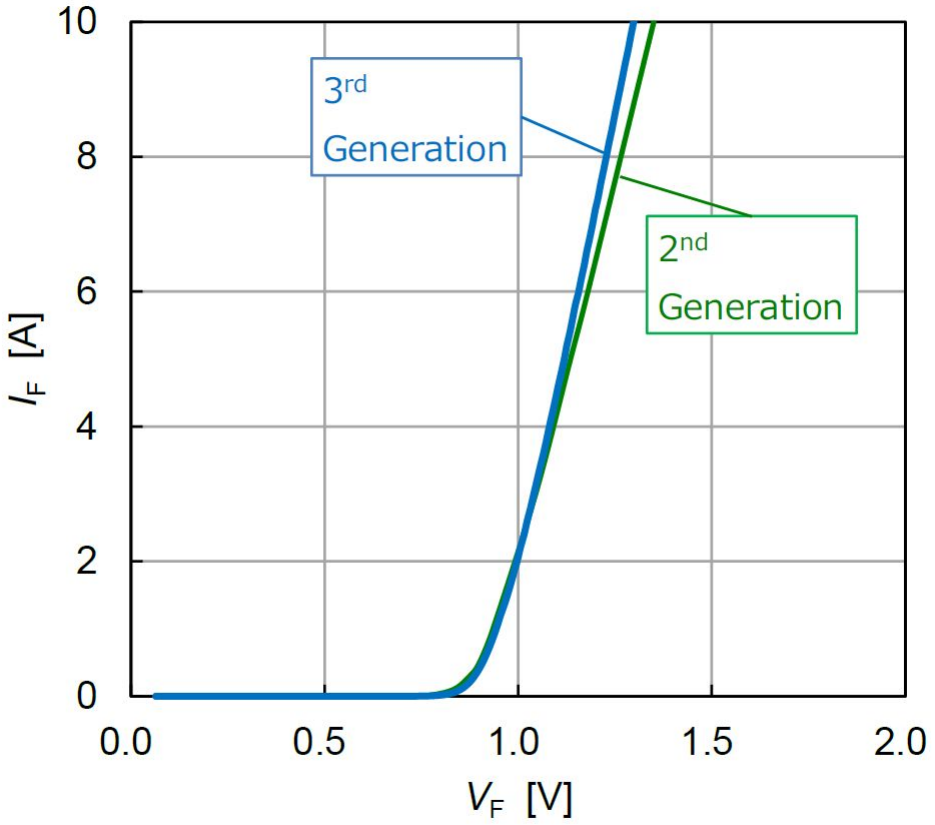
- High breakdown field strength thin drift layer, higher doping concentration low on state loss
- Higher energy gap higher temperature operation (present packaging technology limits operation to 175 ^{o}C)
- Higher thermal conductivity better heat dissipation

High speed switching, high blocking voltage and low on state voltage drop all can be achieved

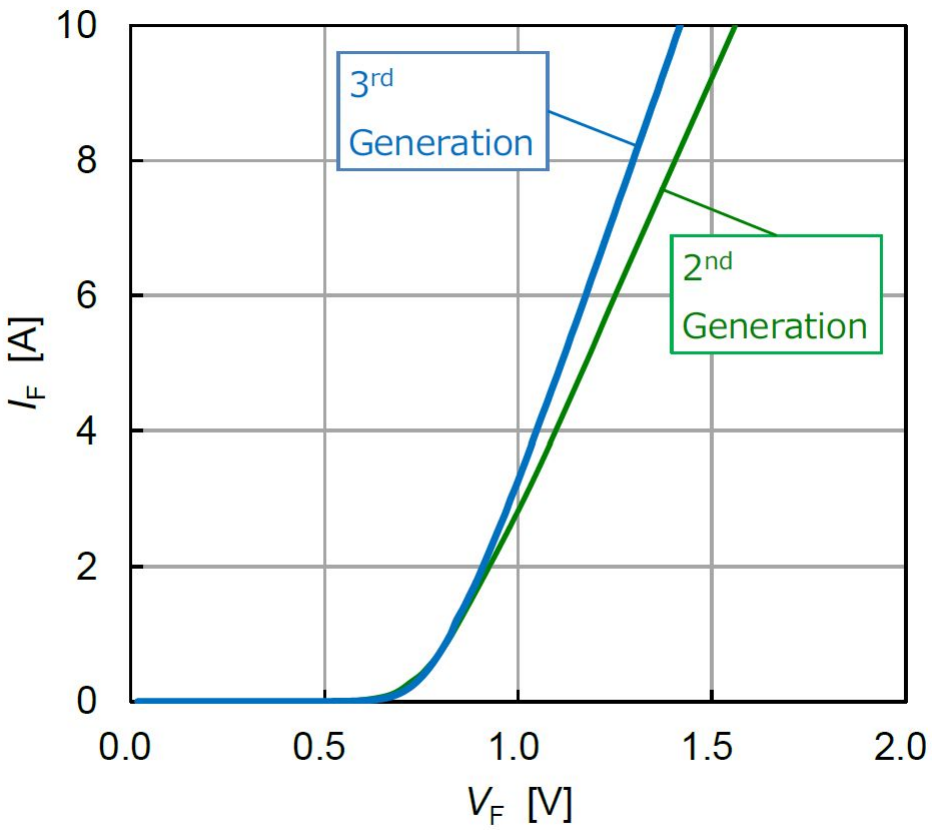
Si and SiC Diodes – A Comparison



Schottky Barrier Diode – Temperature Effect



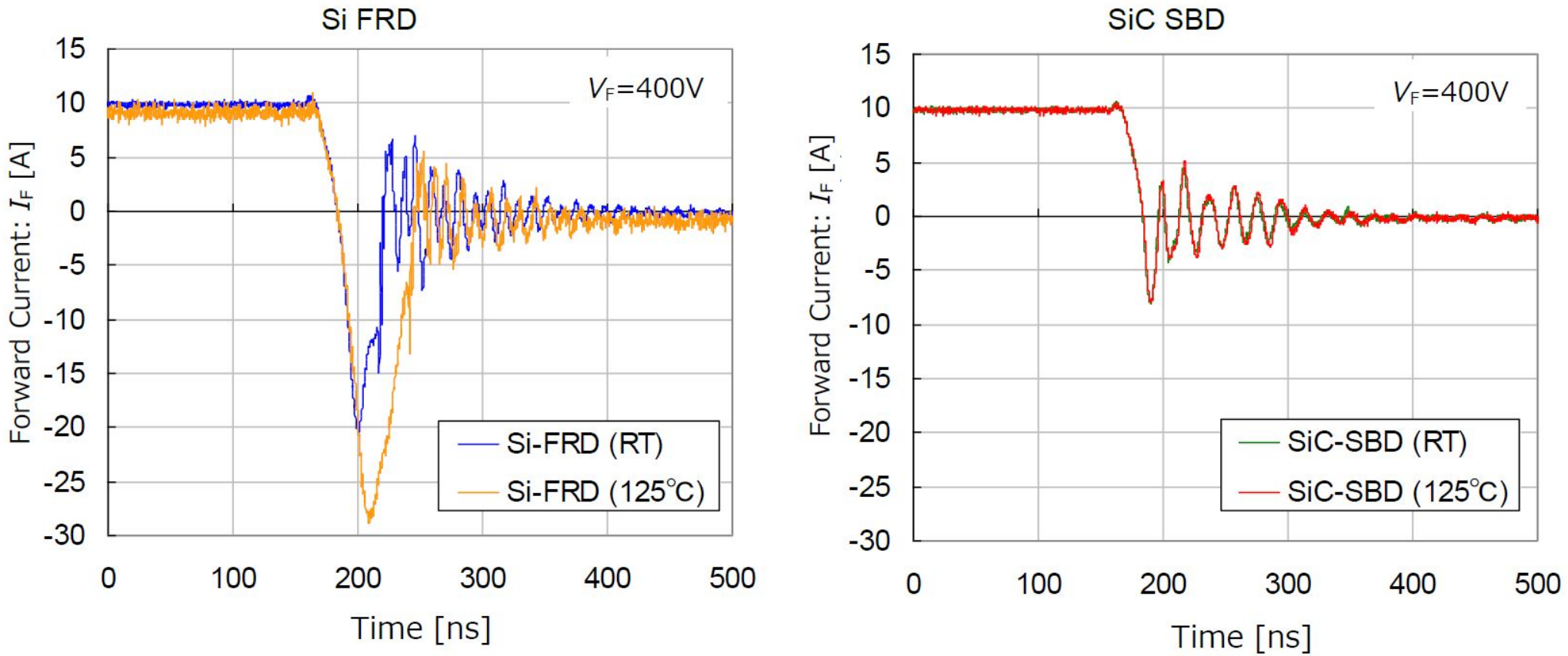
(a) Tj=25℃



(b) Tj=150℃

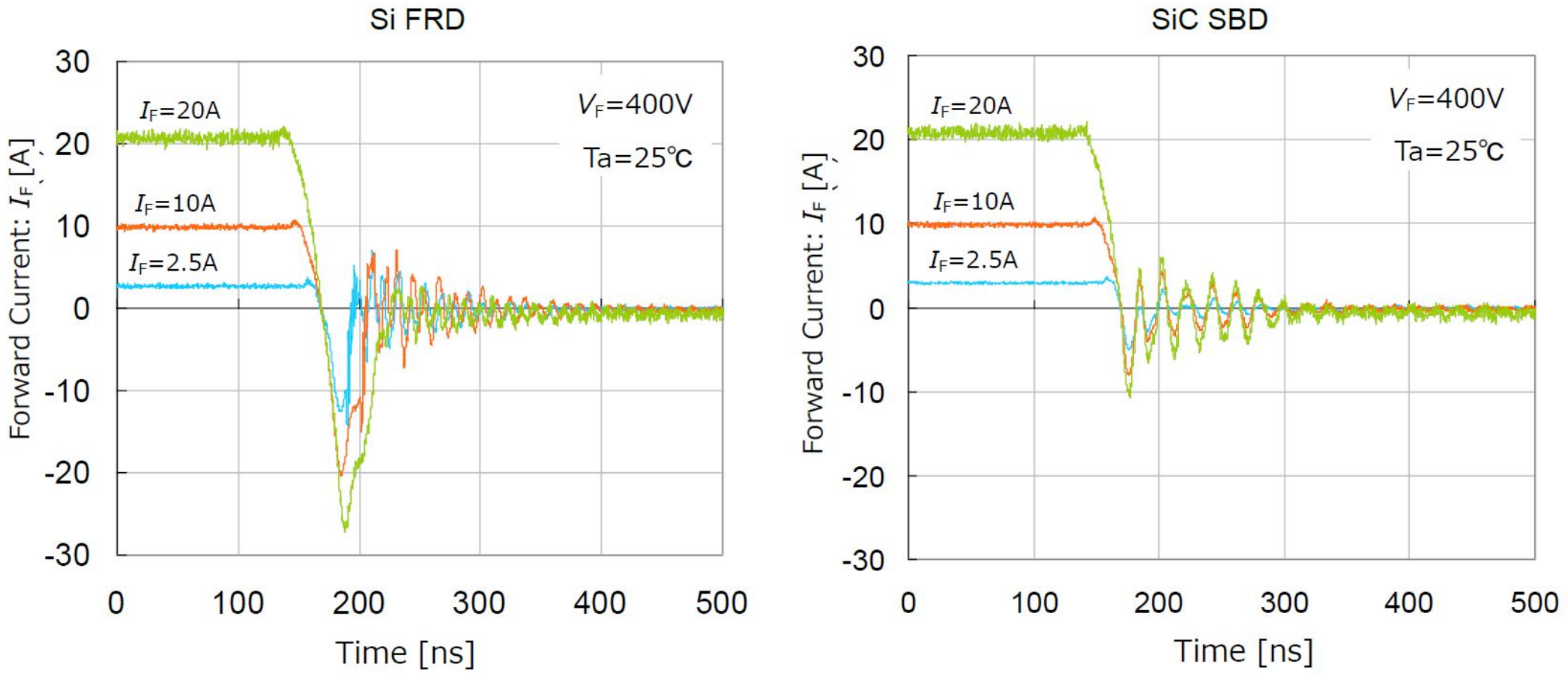
Ease of parallel operation

Temperature Dependence of Diode Reverse Recovery



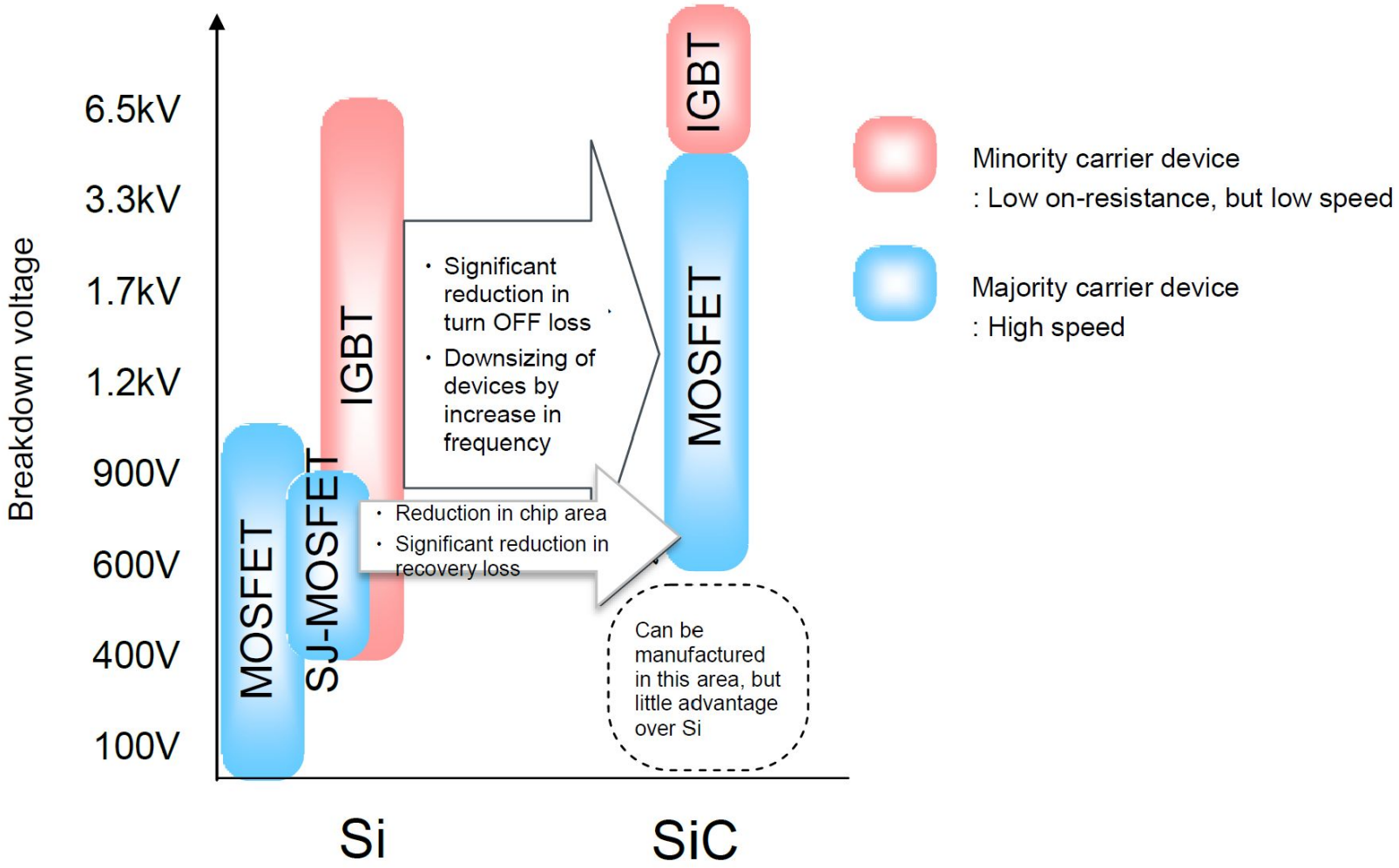
(a) Temperature dependence

Forward Current Dependence of Diode Reverse Recovery

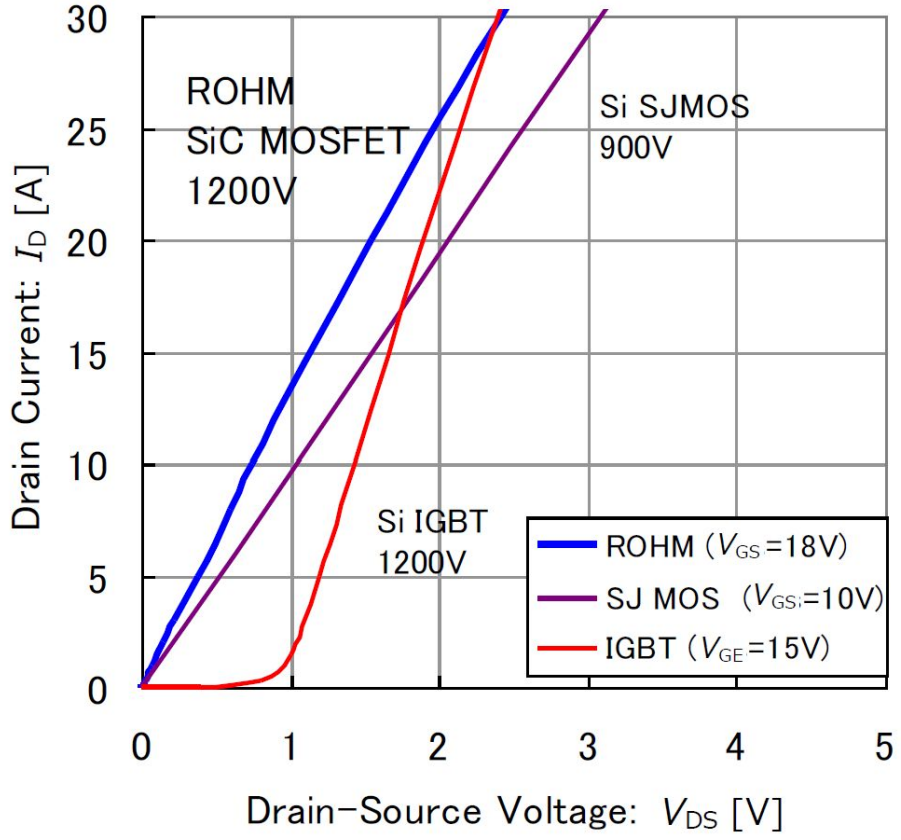


(b) Forward current dependence

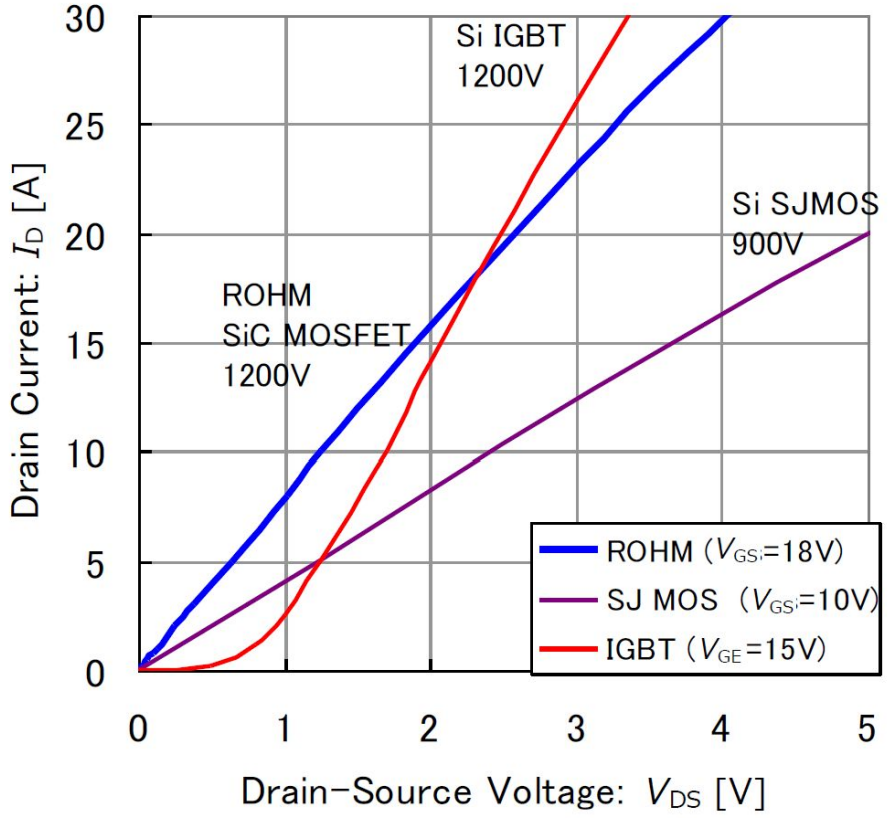
Si and SiC MOSFETs and IGBTs – A Comparison



Si IGBT Vs SiC MOSFET



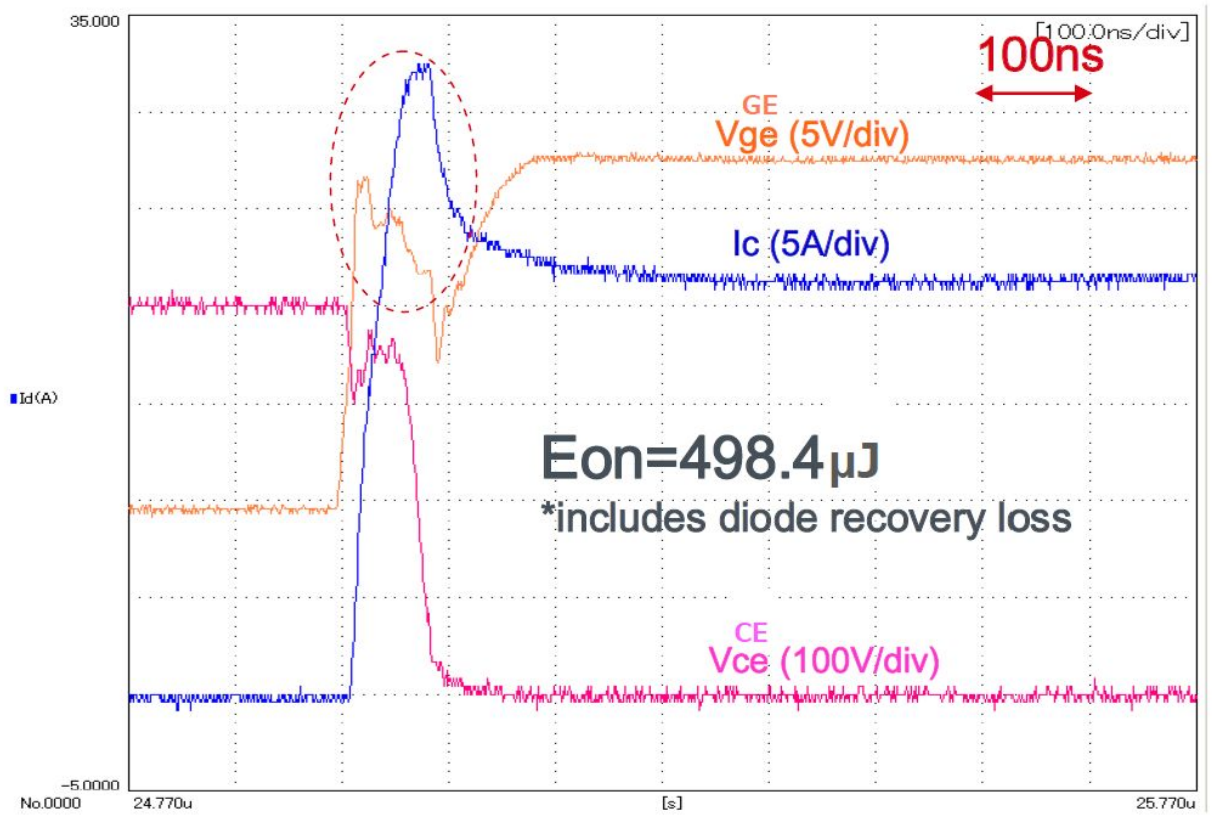
(a) Ta=25℃



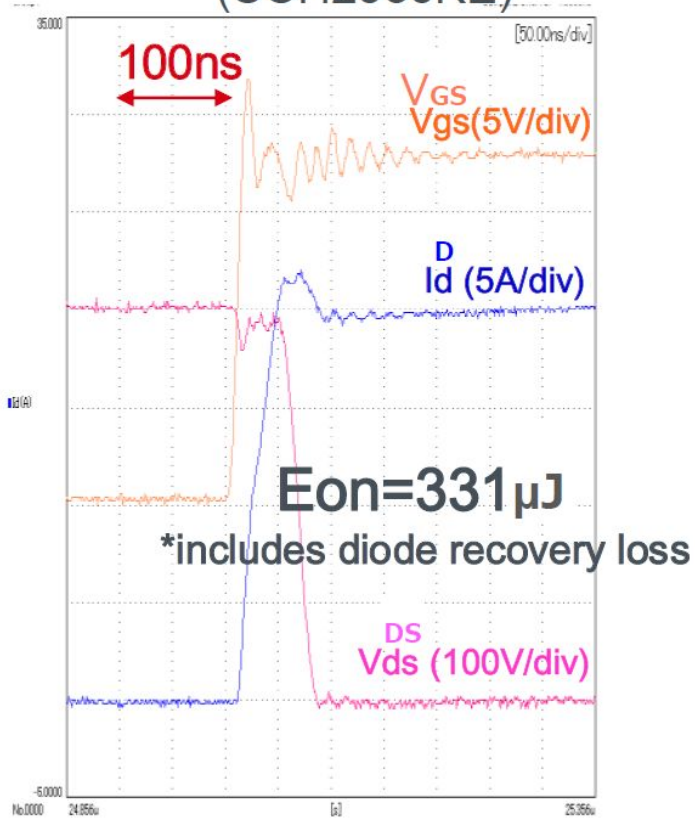
(b) Ta=150℃

Si IGBT Vs SiC MOSFET - Turn On Characteristics

IGBT + FRD

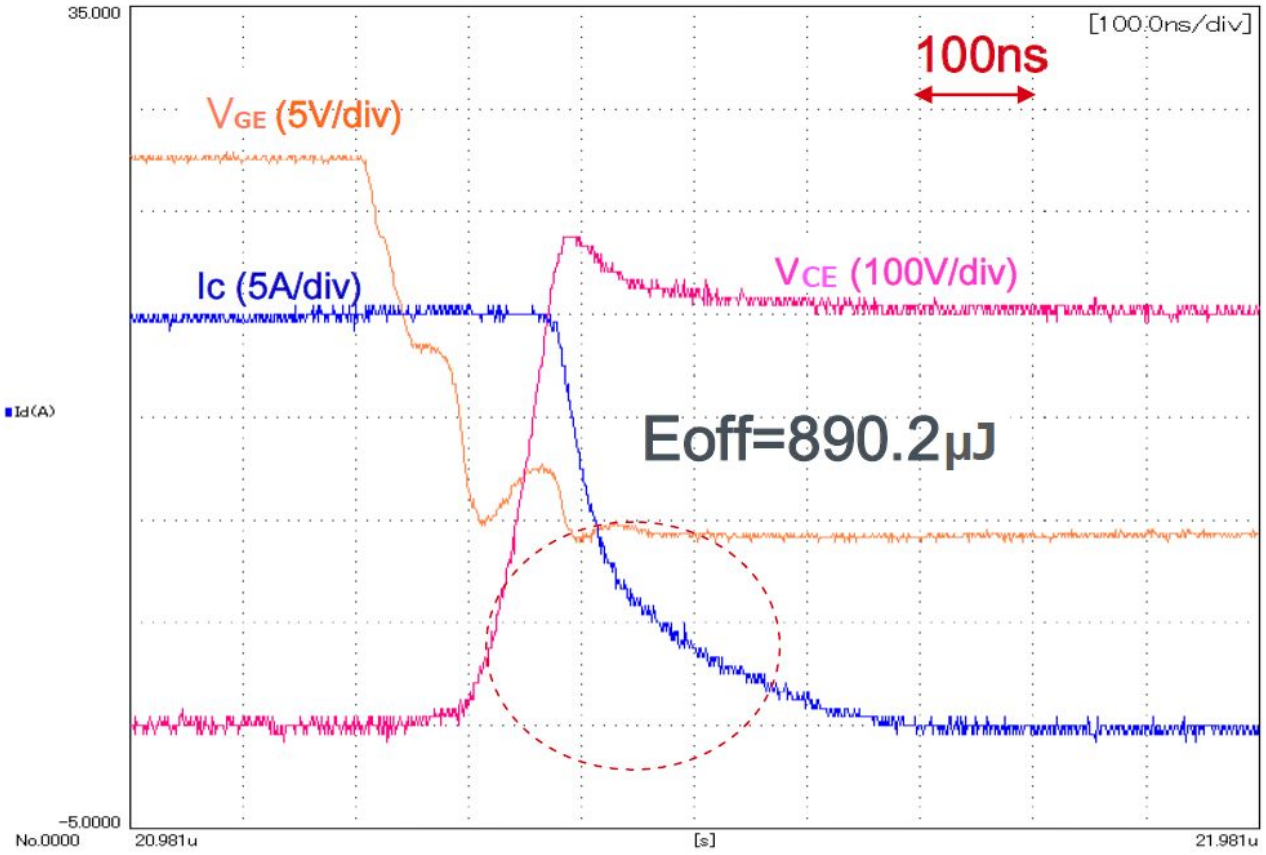


SiC MOSFET + SiC SBD
(SCH2080KE)

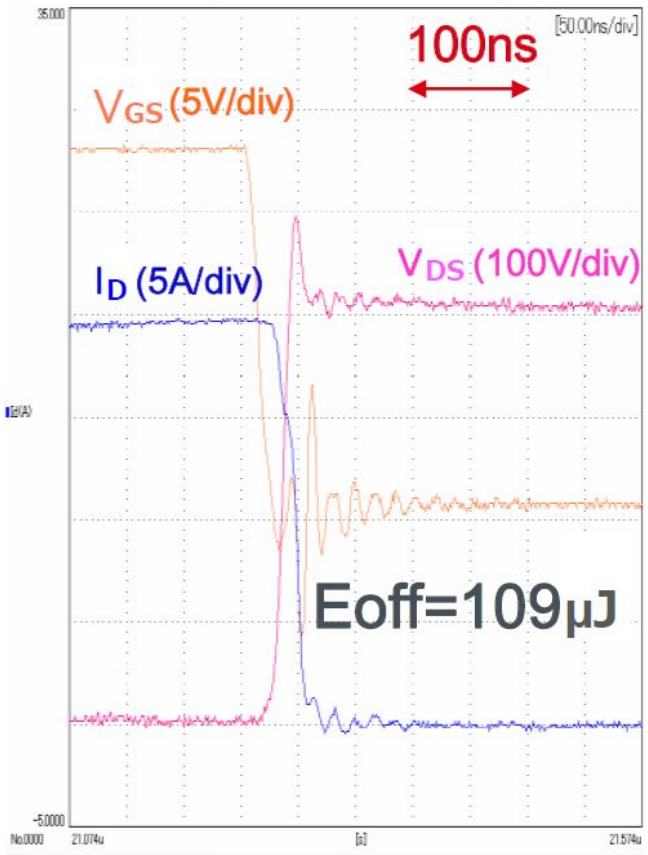


Si IGBT Vs SiC MOSFET – Turn Off Characteristics

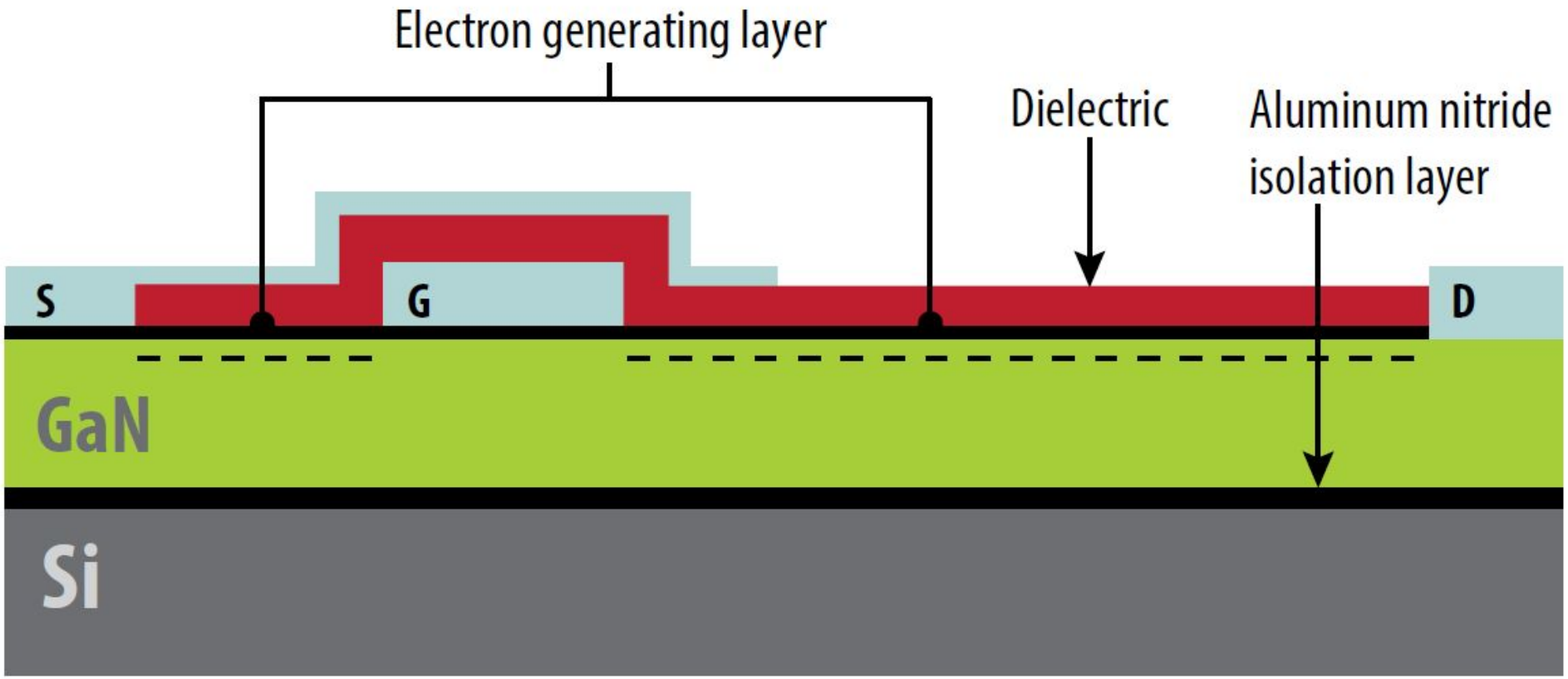
IGBT + FRD



SiC MOSFET + SiC SBD
(SCH2080KE)



GaN E-HEMT Structure



GaN Reverse Conduction

